

Trust the View That Sees the Target: Mining Cross-View Conflicts for Reliability-Gated Disaster Damage Assessment

Yifan Yang*

yyang295@tamu.edu

Department of Geography

Texas A&M University

College Station, Texas, USA

Abstract

After a disaster, building damage is often assessed from two kinds of imagery at once: overhead tiles and ground-level photographs. Most methods fuse the two views symmetrically, trusting both equally for every building. This paper looks instead at the samples where that assumption fails. We call them *conflict cases*: samples on which two independently trained single-view models disagree. We mine such cases from three paired collections, inspection photographs from the 2025 Eaton wildfire and street-view panoramas from Hurricanes Ian and Milton, each matched to very-high-resolution overhead tiles. Conflicts make up 10–33% of each collection. On these samples, an oracle that simply trusts the correct view beats every fusion method we tested by 0.37–0.41 accuracy, and the gap persists after longer training, calibration, and backbone changes. We recover part of it with a visibility-conditioned reliability gate: a linear model that decides which view to trust from building-visibility features, calibrated per-view confidences, and the disagreement itself. On the wildfire data the gate is the only method that significantly beats calibrated probability averaging (+0.051 on conflicts, $p = 0.0001$) and end-to-end fusion (+0.072, $p < 10^{-4}$); on the panoramic datasets it matches them. A controlled field-of-view experiment explains why: cropping each panorama toward the building doubles the benefit of fusion, whereas random crops of the same size do not. Finally, the spatial density of conflicts predicts tile-level damage without any labels (Spearman $r = 0.615$, $p = 0.001$). Mining conflicts turns the question “does fusion help?” into “which view should be trusted, where, and why?”

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Machine learning*; • **Information systems** → *Geographic information systems*.

Keywords

cross-view fusion, street-view imagery, satellite imagery, disaster damage assessment, reliability estimation, GeoAI

*Corresponding author.

ACM Reference Format:

Yifan Yang. 2026. Trust the View That Sees the Target: Mining Cross-View Conflicts for Reliability-Gated Disaster Damage Assessment. In *The 5th ACM SIGSPATIAL International Workshop on Searching and Mining Large Collections of Geospatial Data (GeoSearch '26)*, November 03–06, 2026, Riverside, CA, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3849732.3857333>

1 Introduction

Building-level damage assessment after a disaster increasingly draws on two large image collections at once. Overhead imagery covers a whole region within hours and is the basis of standardized damage mapping [2]. Ground-level imagery, whether crowdsourced street-view panoramas or official inspection photographs, shows the facades and debris that overhead sensors cannot see [8, 10]. Methods that pair the two views are used for geolocalization, damage mapping [4, 5, 9], and multimodal arbitration [7]. Their usual conclusion is encouraging: fusing the views improves average accuracy over either view alone.

We argue that average accuracy hides both the real value and the real limitation of cross-view fusion. For most buildings in a paired collection the two views agree, and fusion changes little. The interesting records are the *conflicts*, the samples on which independently trained single-view models disagree. Conflicts concentrate where one sensor is blind. A facade can look intact from above and gutted from the street; a roof can be destroyed while trees hide it from the road. Finding these records in a large collection, deciding which view to believe for each one, and reading their spatial pattern is as much a search-and-mining problem as a modeling problem.

We therefore make conflict cases the unit of analysis and ask three questions. First, how much information do fusion methods leave unused on conflicts? We measure an *oracle single-view gap*: the margin between an oracle that trusts whichever view is correct for each sample and the best available method. Across three disasters this gap is 0.37–0.41 accuracy, several times larger than any fusion gain. Second, can the gap be recovered, and how? We find that per-sample view reliability is largely predictable from whether the ground image actually *sees the target building*. A linear gate over segmentation-derived visibility, calibrated confidence, and disagreement features recovers part of the gap and can be read as a one-sentence rule. Third, is the effect causal? Datasets differ in hazard, geography, sensors, and labels all at once, so we manipulate target alignment *within* a dataset with a controlled cropping experiment. A fourth result is directly useful for mining post-disaster collections: the spatial density of conflicts is an unsupervised damage map that can be computed before any labels



This work is licensed under a Creative Commons Attribution 4.0 International License. *GeoSearch '26, Riverside, CA, USA*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-3058-0/2026/11

<https://doi.org/10.1145/3849732.3857333>

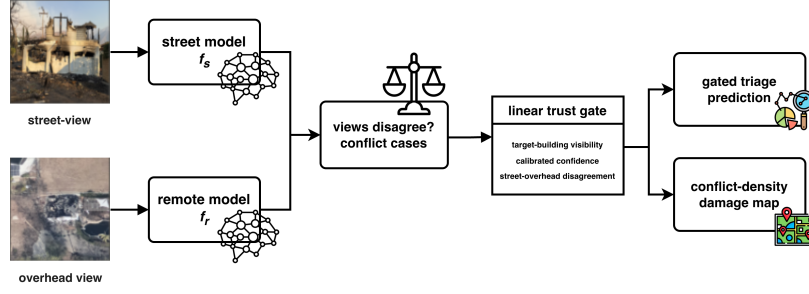


Figure 1: Overview. Independently trained single-view models define the conflict cases. A linear gate over target-building visibility, calibrated confidence, and cross-view disagreement decides which view to trust for each sample, and the spatial density of conflicts doubles as a label-free damage map. The pair shown is a real Eaton-fire conflict case.

Table 1: Test-set results (five-seed means; accuracy std ≤ 0.03 everywhere). Conflict accuracy is computed on the per-seed conflict subset; closure follows Eq. (1) with the oracle at 0.961 / 0.896 / 0.964 conflict accuracy (gap 0.410 / 0.367 / 0.407). Gate = three-view linear reliability gate.

Method	Eaton wildfire (conflict rate 9.6%)			Hurricane Ian (33.2%)			Hurricane Milton (25.7%)		
	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure
street_only	0.931	0.486	—	0.703	0.529	—	0.759	0.557	—
remote_only	0.929	0.475	—	0.648	0.367	—	0.721	0.407	—
concat	0.930	0.674	0.29	0.709	0.588	0.15	0.766	0.622	0.16
crossview (end-to-end fusion)	0.940	0.699	0.35	0.735	0.624	0.25	0.776	0.649	0.22
calibrated probability averaging	0.954	0.732	0.44	0.731	0.617	0.24	0.776	0.626	0.17
reliability gate (linear)	0.959	0.768	0.52	0.731	0.618	0.23	0.777	0.631	0.18

exist (Figure 1). Code, protocols, and derived tables are available at <https://github.com/rayford295/CrossViewGate>.

2 Data and Protocol

We use three paired street/overhead collections. They span two hazards and, more importantly, two ground-view *observation regimes*: property-centric photographs that are aimed at a building, and panoramas that are not.

Eaton wildfire (Altadena, CA; property-centric). Building-level CAL FIRE Damage Inspection (DINS) photographs taken after the January 2025 fire, paired by location with post-event VHR overhead tiles. The six-level DINS scale is mapped to three ordinal classes (no/trace damage, damaged–repairable, destroyed). The split has 6.5k/1,966/1,984 train/val/test pairs. **Hurricane Ian (panoramic).** The CVIAN pairing released with CVDIsaster [5]: Mapillary 360° panoramas over Sanibel Island, 27 cm post-event satellite imagery, and three-level damage-perception labels, with 3,248/573/300 pairs. **Hurricane Milton (panoramic).** A 360° street-view/satellite pairing over Florida after the October 2024 hurricane, built in the same way; 1,740/307/254 pairs.

Splits and leakage. Eaton is split by property and tile, so all photographs of one structure fall in one partition. For Ian we keep the released split for the headline numbers, so that they are comparable with CVDIsaster. However, an audit against the official CVIAN position file found that 299 of the 300 test samples share a Mapillary sequence with the training data. We therefore also report a repaired protocol with 0.005° spatial blocks and a 25 m buffer (Section 4).

3 Method

Base models and conflict cases. For each dataset we train four models under one protocol: *street_only* and *remote_only* (ResNet-18 [3] with a linear head), *concat* (both encoders with an MLP head), and *crossview* (both encoders with a learned interaction head), which serves as the end-to-end fusion reference. Training uses class-balanced cross-entropy, 224-pixel inputs, AdamW, up to 15 epochs with early stopping on validation macro-F1, and five seeds. A sweep over ResNet-50, CLIP, and DINOv2 backbones changed no conclusion. Writing $s(x)$ and $r(x)$ for the two single-view predictions, the **conflict subset** is $C = \{x : s(x) \neq r(x)\}$. Conflict rates are 9.6% on Eaton, 33.2% on Ian, and 25.7% on Milton.

Oracle single-view gap. The oracle counts a conflict sample as correct if *either* single view is correct. For a method m we report its conflict accuracy $\text{acc}_C(m)$ and its **oracle-gap closure**,

$$\text{closure}(m) = \frac{\text{acc}_C(m) - \text{acc}_C(\text{best single})}{\text{acc}_C(\text{oracle}) - \text{acc}_C(\text{best single})}. \quad (1)$$

The oracle only chooses *which* of two fixed models to trust. It bounds what arbitration between existing predictors can achieve, not what new models could.

Calibration. Averaging miscalibrated probabilities can win or lose for reasons unrelated to information content [1]. We therefore fit one temperature per view on validation predictions and evaluate every training-free fusion baseline on calibrated probabilities. Temperature scaling does not change the argmax, so the conflict subset is unchanged.

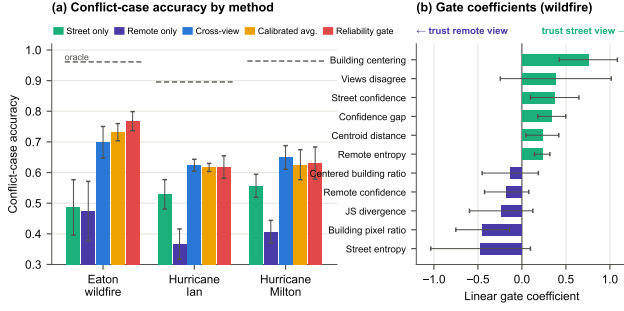


Figure 2: (a) Conflict-case accuracy by method (mean $\pm$ std over five seeds; dashed line = oracle). (b) Linear gate coefficients on the wildfire (mean $\pm$ std over seeds); positive values shift trust toward the street view.

Visibility-conditioned reliability gate. A frozen SegFormer-B0 [6] trained on ADE20K gives four building-visibility features per street image: the building pixel ratio, the same ratio inside a centered crop of half the image size, their difference (a centering signal), and the normalized distance of the building centroid from the image center. Seven more features describe the predictions themselves: calibrated confidence and entropy for each view, the confidence gap, the Jensen–Shannon divergence between the two calibrated distributions, and a disagreement flag. From these eleven features the gate outputs softmax weights over the street, remote, and crossview probabilities. It is fit on the validation split by minimizing the negative log-likelihood of the gated mixture, with z-scored features. The headline gate is *linear*. A small MLP matched it throughout, and linear coefficients can be read directly.

Field-of-view intervention. Each 1024×512 panorama is cropped to a 256×256 window with a 90° horizontal field of view on the horizon band. In the **building-centered** variant the window is aimed at the circular-mean azimuth of the SegFormer building mask. In the **random** variant it is aimed at a reproducible uniformly random azimuth. The two variants share size, projection, and information budget; they differ only in whether the window points at the built structure. We retrain *street_only* and *crossview* on each variant with three seeds, reuse the unchanged remote models, and compare the gain of fusion on conflicts.

Conflict density. We aggregate seed-averaged predictions to geographic tiles. For each tile we compute the hard conflict density (the fraction of samples on which the two argmax predictions disagree) and, as a control, the mean predictive entropy of the single-view models. Both are correlated (Spearman) with the tile’s mean ordinal damage.

Statistics. Rather than reporting “significant in k of N seeds”, we average each sample’s paired correctness difference over the seeds in which it qualifies, and test whether the mean difference is zero with a two-sided sign-flip permutation test (20,000 permutations) and percentile bootstrap 95% intervals.

Table 2: Field-of-view intervention (three seeds; remote models unchanged). Conflict gain = crossview – best single view on conflicts.

Dataset	Street-view variant	Street acc.	Conflict gain	Closure
Ian	original panorama	0.709	+0.064	0.166
Ian	building-centered crop	0.648	+0.109	0.269
Ian	random crop (control)	0.574	+0.042	0.072
Milton	original panorama	0.740	+0.067	0.177
Milton	building-centered crop	0.693	+0.149	0.365
Milton	random crop (control)	0.654	+0.055	0.128

4 Results

Overall results and the oracle gap. Table 1 shows three patterns. First, cross-view fusion beats both single views on conflicts on every dataset, and the pooled tests confirm it (crossview minus street: Eaton +0.246, $p < 10^{-4}$; Ian +0.078, $p = 0.013$; Milton +0.141, $p = 0.006$). Second, once the single-view models are converged and calibrated, simple probability averaging is a strong baseline. On Eaton it exceeds plain crossview (0.954 vs. 0.940), and end-to-end fusion does not significantly beat it on any dataset (Eaton conflicts +0.034, $p = 0.073$). Reports that learned fusion beats simple aggregation may therefore reflect undertrained or miscalibrated baselines. Third, the oracle gap is much larger than any of these differences. Knowing *which existing model to trust* for each sample would add 0.37–0.41 conflict accuracy. Calibration does not explain the gap, since calibrated averaging closes at most 44% of it. The unused information concerns *when* each view is reliable.

The reliability gate. On the property-centric dataset the gate behaves as the mechanism predicts (Figure 2a). It gains +0.072 conflict accuracy over end-to-end fusion (95% CI [+0.046, +0.097], $p < 10^{-4}$), +0.018 on the full test set ($p < 10^{-4}$), and +0.051 over calibrated averaging ($p = 0.0001$). Its closure of 52% is the only one above one half. On the panoramic datasets the street views rarely see the target: the mean building pixel ratio is 0.027 on Ian against 0.119 on Eaton. There the gate has little visibility signal to use and matches crossview (Ian +0.005, $p = 0.70$; Milton −0.015, $p = 0.57$). This asymmetry is the central finding. Gating pays where ground evidence is aimed at the target, and it costs nothing where it is not. The gate coefficients (Figure 2b) keep their signs across seeds. Street entropy has the largest negative weight, while building centering and the confidence gap are positive. In words, the learned rule is: *trust the street view when it is confident and is actually looking at the building*.

Field-of-view intervention. Building-centered cropping raises the conflict gain on both hurricanes (Table 2). On Milton the closure doubles, from 0.177 to 0.365, which is two-thirds of the wildfire level. Random cropping with identical geometry lowers the gain instead. The building-minus-random contrast is positive in all six dataset-seed pairs. Note that building-centered cropping *lowers* street-only accuracy yet *raises* the benefit of fusion. The gain therefore does not come from a stronger street model; it comes from the two views attending to the same object, and the random control rules out cropping artifacts. Target alignment of the street view is the causal variable behind the regime effect, which supports reading the gate’s visibility features as causal rather than merely correlated.

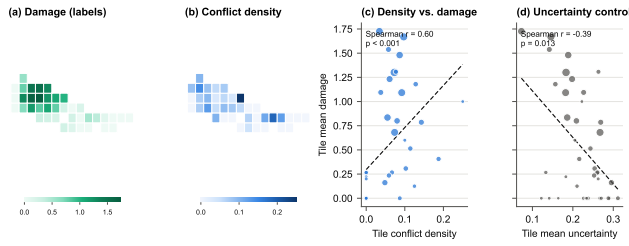


Figure 3: Conflict density as a label-free damage map (Eaton wildfire). (a) Labeled tile damage. (b) Tile conflict density computed without labels. (c) Their rank correlation. (d) The single-view uncertainty control, which anti-correlates with damage. Point size reflects tile sample count.

Conflict density as a spatial damage signal. On the wildfire, the conflict density of an overhead tile is a strong label-free predictor of its damage (Spearman $r = 0.615$, $p = 0.001$ over 25 tiles; $r = 0.60$ on a 0.01° grid; Figure 3). The correlation grew as the models converged (from 0.545 to 0.615). The uncertainty control behaves very differently: it *anti-correlates* with damage ($r = -0.37$ to -0.52), because parcels burned to the ground are easy, confident classifications. Conflict density is therefore not model uncertainty under another name. It is the spatial signature of two sensors disagreeing about the world. Damage on Eaton is spatially autocorrelated (Moran’s $I = 0.27$, $p = 0.0001$), so part of the tile correlation reflects spatial structure. On Milton the signal drops below significance once the models converge ($r = 0.289$), because better single-view models disagree less. The wildfire result is the robust one.

External comparison and split repair. On the same CVIAN pairing, CVDDisaster’s CGCViT [5] reports 77.96 overall accuracy with a 20M-parameter encoder under a random split; our ResNet-18 crossview under the released split reaches 73.5 ± 1.5 . Under the repaired spatial-block split, macro-F1 drops by 0.05–0.10 for every model. Random or index-based splits thus overstate spatial generalization. The single-view ranking (street above satellite) is stable, but the size of the cross-view advantage depends on the protocol.

5 Discussion and Conclusion

The 0.37–0.41 oracle gap measures the opportunity within a restricted choice between two existing single-view models, and it is far larger than what architectural work on symmetric fusion has produced. For mining paired geospatial collections, the productive question is therefore not “how should features be fused?” but “when should each view be believed?”. The gate closes half of the gap on property-centric imagery with eleven interpretable features; the remainder is a quantified target for richer visibility descriptors and pre-event reference imagery. In practice the gate needs one frozen segmentation pass per street image, a temperature, a linear fit, and the output probabilities of existing models. Its rule fits in one sentence, which matters where automated triage must be auditable. Where ground imagery is panoramic and rarely aimed at the target, calibrated averaging captures most of the achievable benefit. Where it is property-centric, as in inspection programs and insurance photography, gating pays. Because the driver is alignment rather than

hazard, capture policy is itself an intervention: aiming cameras at structures raises the value of every downstream fusion component.

Limitations. The property-centric regime rests on one wildfire dataset. Milton is small (2,301 pairs). The repaired CVIAN test set has six spatial blocks linked by shared Mapillary sequences, so strong spatial-generalization claims need more groups. The ADE20K segmenter measures built-structure visibility, not *the* target building. The density signal needs replication on further geo-referenced disasters.

Conclusion. On the samples where the two views disagree, symmetric cross-view fusion leaves 0.37–0.41 accuracy on the table across three real disasters. A linear, visibility-conditioned gate recovers a significant part of it; a field-of-view experiment shows that the mechanism, target alignment, is causal; and spatially aggregated disagreement is a label-free damage map. Mining conflicts turns “does fusion help?” into “which view should be trusted, where, and why?”.

Acknowledgments

The author used Claude (Anthropic) to polish the wording and check the grammar of this manuscript. All ideas, experiments, analyses, and claims are the author’s own; the author reviewed every sentence and takes full responsibility for the content.

References

- [1] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*. PMLR, 1321–1330.
- [2] Ritwik Gupta, Bryce Goodman, Nirav Patel, Ricky Hosfelt, Sandra Sajeev, Eric Heim, Jigar Doshi, Keane Lucas, Howie Choset, and Matthew Gaston. 2019. Creating xBD: A Dataset for Assessing Building Damage from Satellite Imagery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. 10–17.
- [3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 770–778.
- [4] Hao Li, Fabian Deuser, Wenping Yin, Steffen Knoblauch, Wufan Zhao, Filip Biljecki, Yong Xue, and Wei Huang. 2026. Towards Generative Location Awareness for Disaster Response: A Probabilistic Cross-View Geolocalization Approach. *ISPRS Journal of Photogrammetry and Remote Sensing* 237 (2026), 130–145.
- [5] Hao Li, Fabian Deuser, Wenping Yin, Xuanshu Luo, Paul Walther, Gengchen Mai, Wei Huang, and Martin Werner. 2025. Cross-View Geolocalization and Disaster Mapping with Street-View and VHR Satellite Imagery: A Case Study of Hurricane Ian. *ISPRS Journal of Photogrammetry and Remote Sensing* 220 (2025), 841–854.
- [6] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. 2021. SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers. In *Advances in Neural Information Processing Systems*, Vol. 34. 12077–12090.
- [7] Yifan Yang, Lei Zou, Wenjing Gong, Kani Fu, Zongrong Li, Siqin Wang, Bing Zhou, Heng Cai, and Hao Tian. 2026. DamageArbiter: A CLIP-Enhanced Multimodal Arbitration Framework for Hurricane Damage Assessment from Street-View Imagery. *arXiv preprint arXiv:2603.14837* (2026).
- [8] Yifan Yang, Lei Zou, Bing Zhou, Daoyang Li, Binbin Lin, Joynal Abedin, and Mingzheng Yang. 2025. Hyperlocal Disaster Damage Assessment Using Bi-Temporal Street-View Imagery and Pre-Trained Vision Models. *Computers, Environment and Urban Systems* 121 (2025), 102335.
- [9] Wenping Yin, Fabian Deuser, Ziqi Liu, Jiaqi Wei, Xuanshu Luo, Martin Werner, Hao Li, and Yong Xue. 2026. Triple-Objective Cross-View Geolocalization of Disaster-Related VGI: The Case of Hurricane Ian. *International Journal of Geographical Information Science* 40, 1 (2026), 194–216.
- [10] Wei Zhai and Zhong-Ren Peng. 2020. Damage Assessment Using Google Street View: Evidence from Hurricane Michael in Mexico Beach, Florida. *Applied Geography* 123 (2020), 102252.